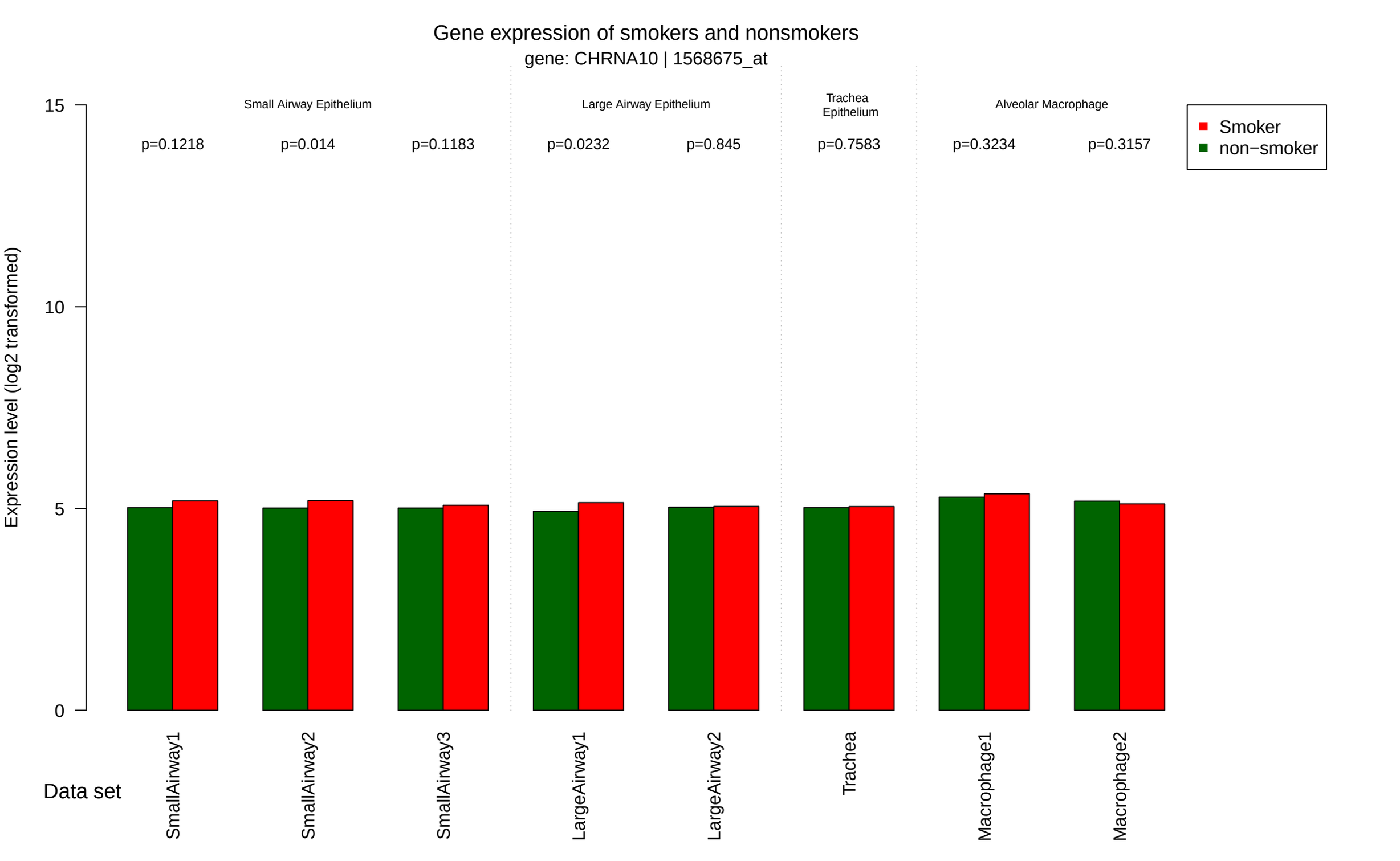
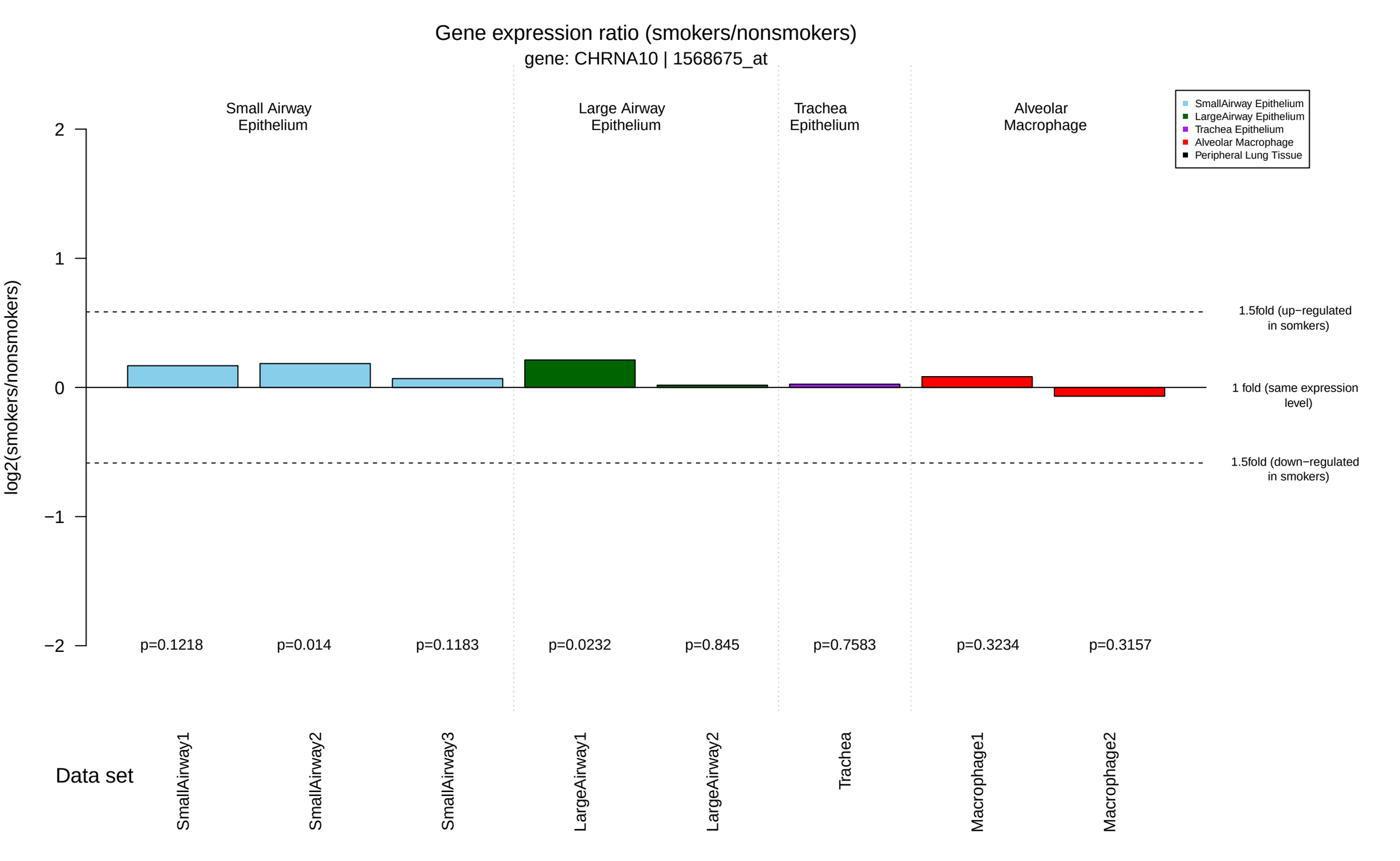
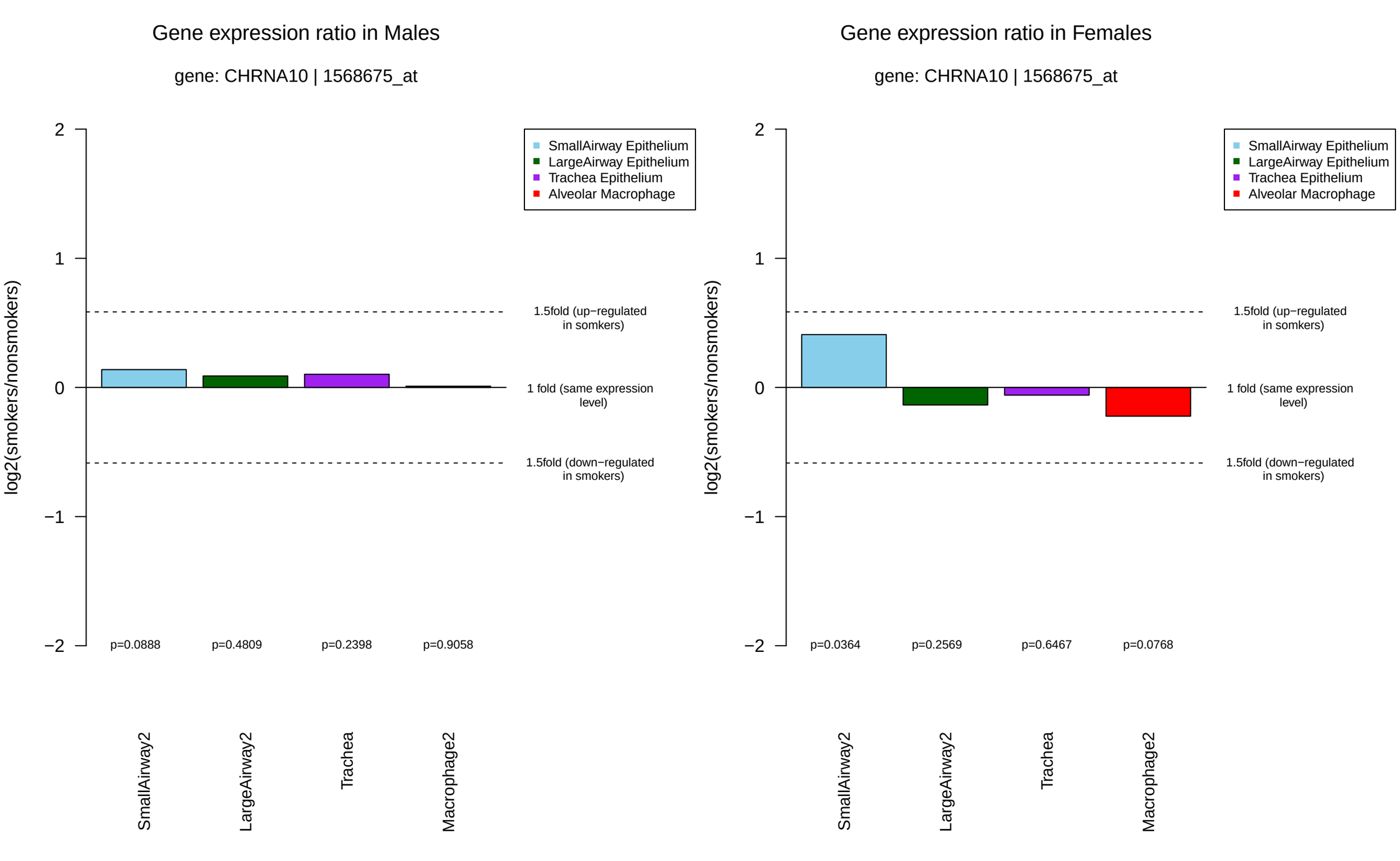
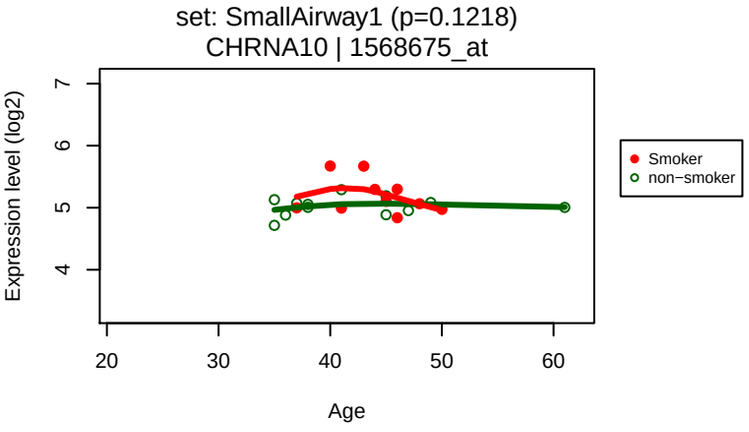
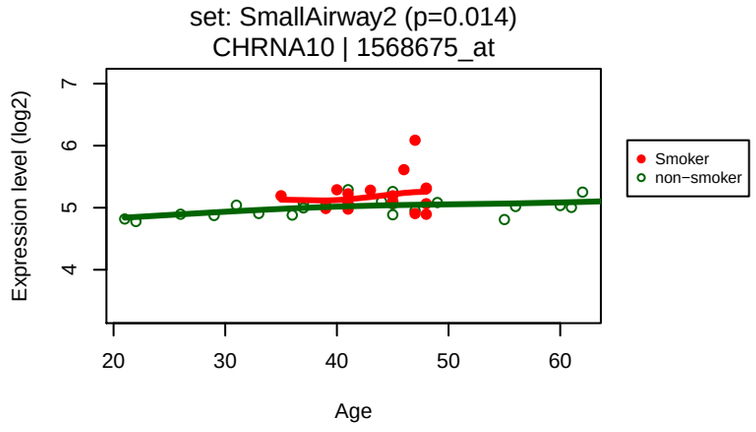
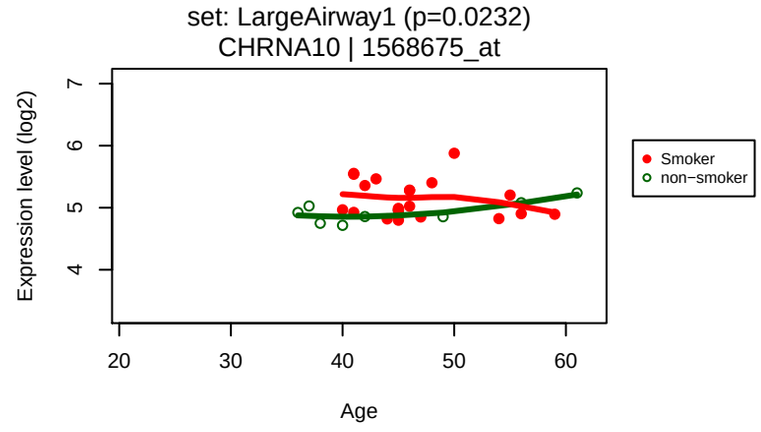

Gene expression of smokers and nonsmokers
gene: CHRNA10 | 1568675_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1218
p=0.014
p=0.1183
p=0.0232
p=0.845
p=0.7583
p=0.3234
p=0.3157
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

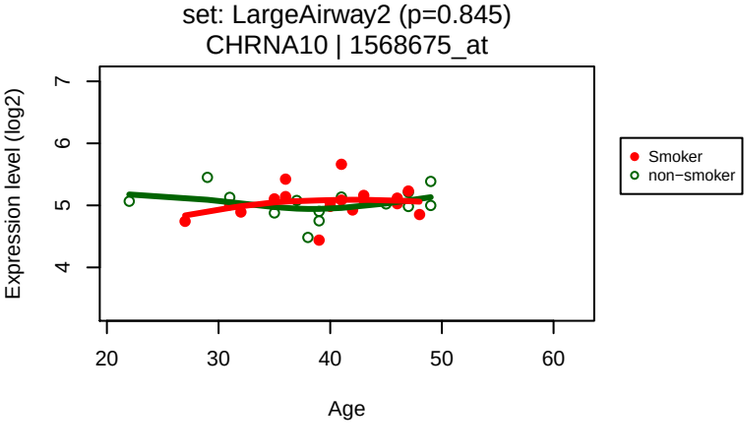

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA10 | 1568675_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1218
p=0.014
p=0.1183
p=0.0232
p=0.845
p=0.7583
p=0.3234
p=0.3157
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

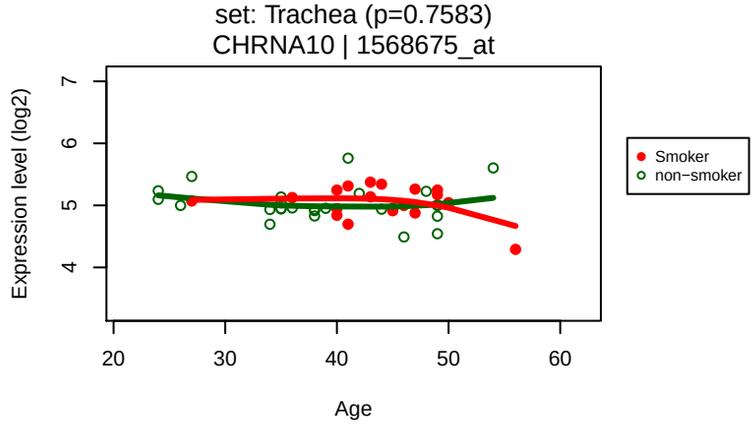

Gene expression ratio in Males
gene: CHRNA10 | 1568675_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0888
p=0.4809
p=0.2398
p=0.9058
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA10 | 1568675_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0364
p=0.2569
p=0.6467
p=0.0768
SmallAirway2
LargeAirway2
Trachea
Macrophage2

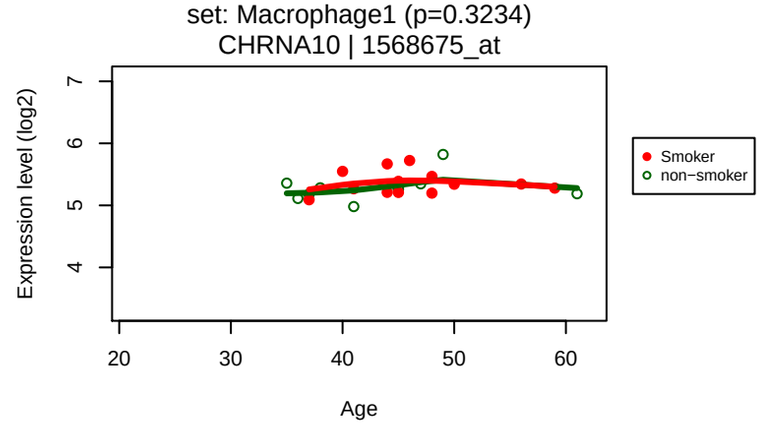

set: SmallAirway1 (p=0.1218)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker

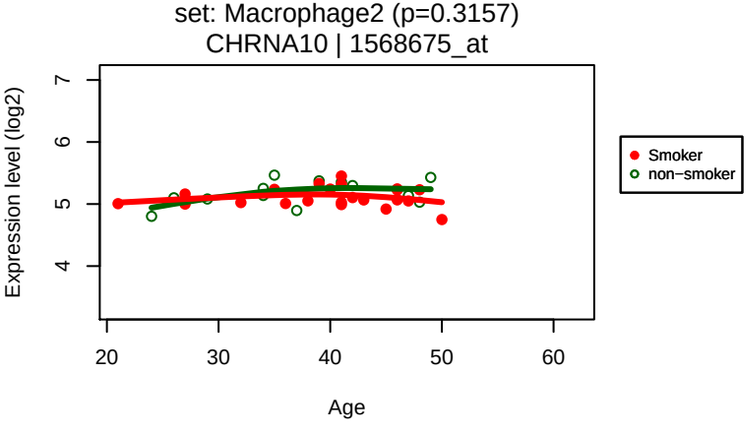

set: SmallAirway2 (p=0.014)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker

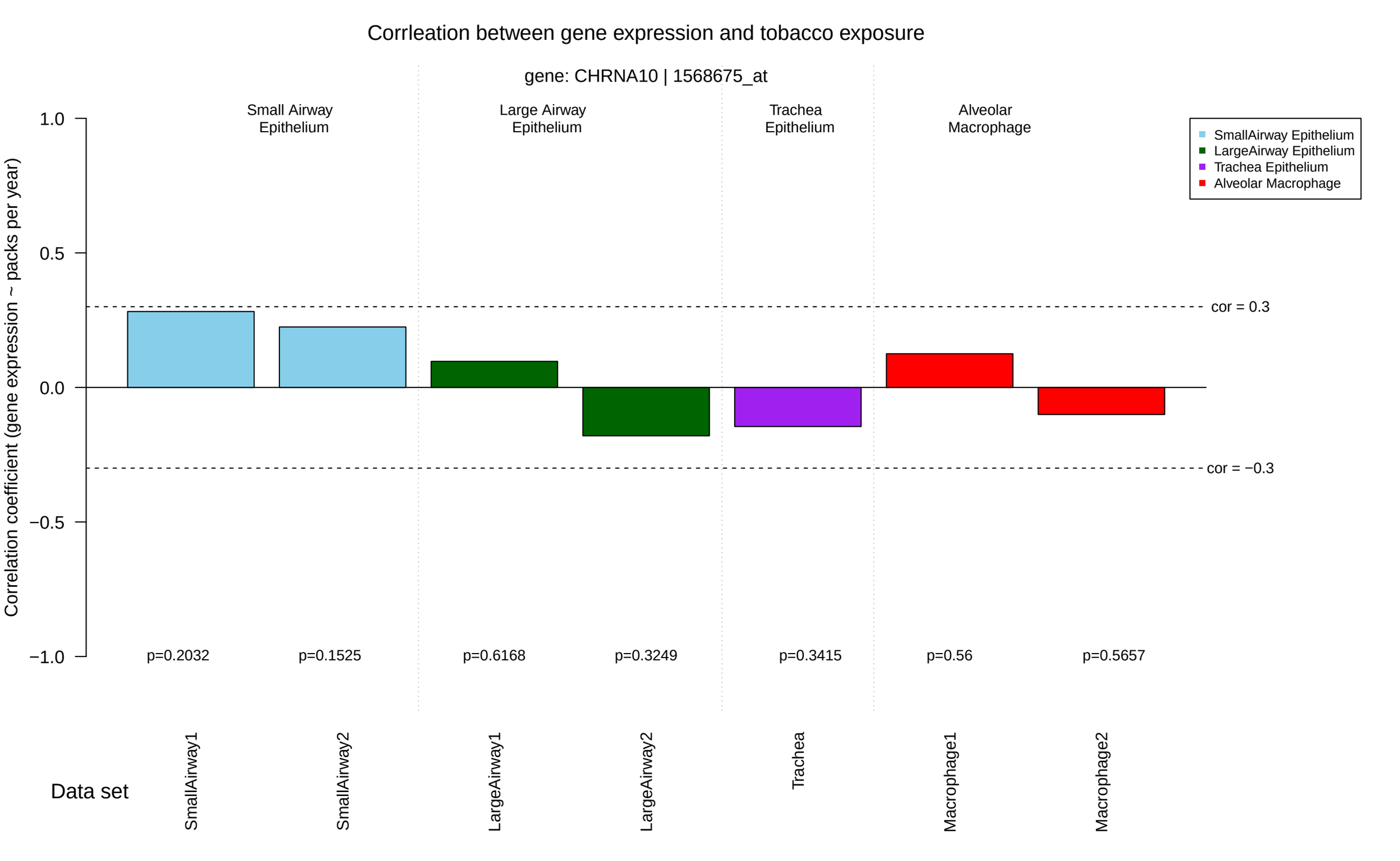

set: LargeAirway1 (p=0.0232)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.845)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7583)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3234)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3157)
CHRNA10 | 1568675_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA10 | 1568675_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2032
p=0.1525
p=0.6168
p=0.3249
p=0.3415
p=0.56
p=0.5657
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2